INCH-POUND

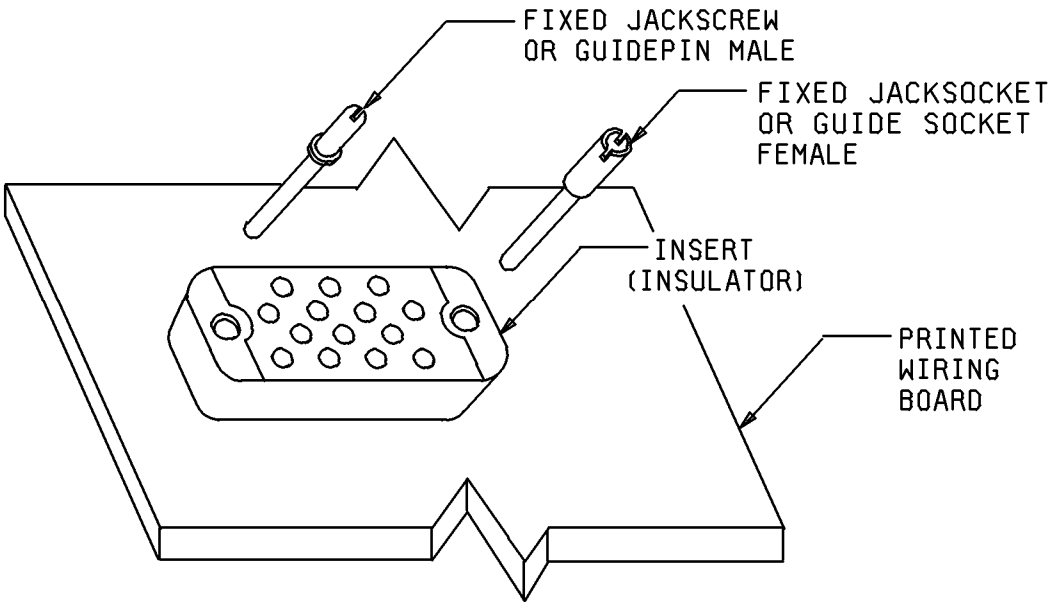
MIL-DTL-28748/19
11 March 2005

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR,
RACK AND PANEL, PRINTED WIRING BOARD (PWB) MOUNT,
STRAIGHT THROUGH, SOCKET CONTACTS, SIZE 22

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28748.

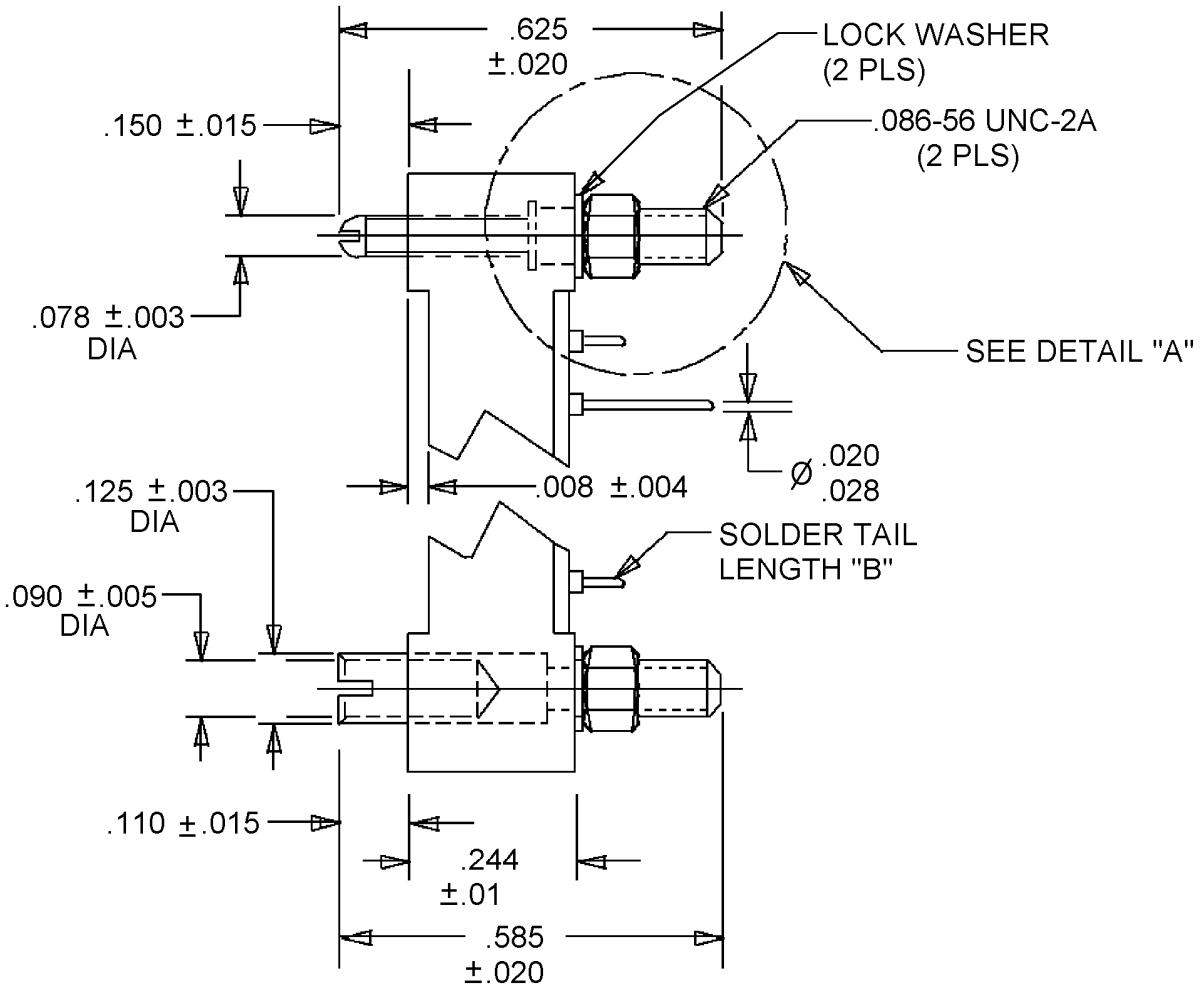



Connector

FIGURE 1. Connector with jackscrews and guidepins.



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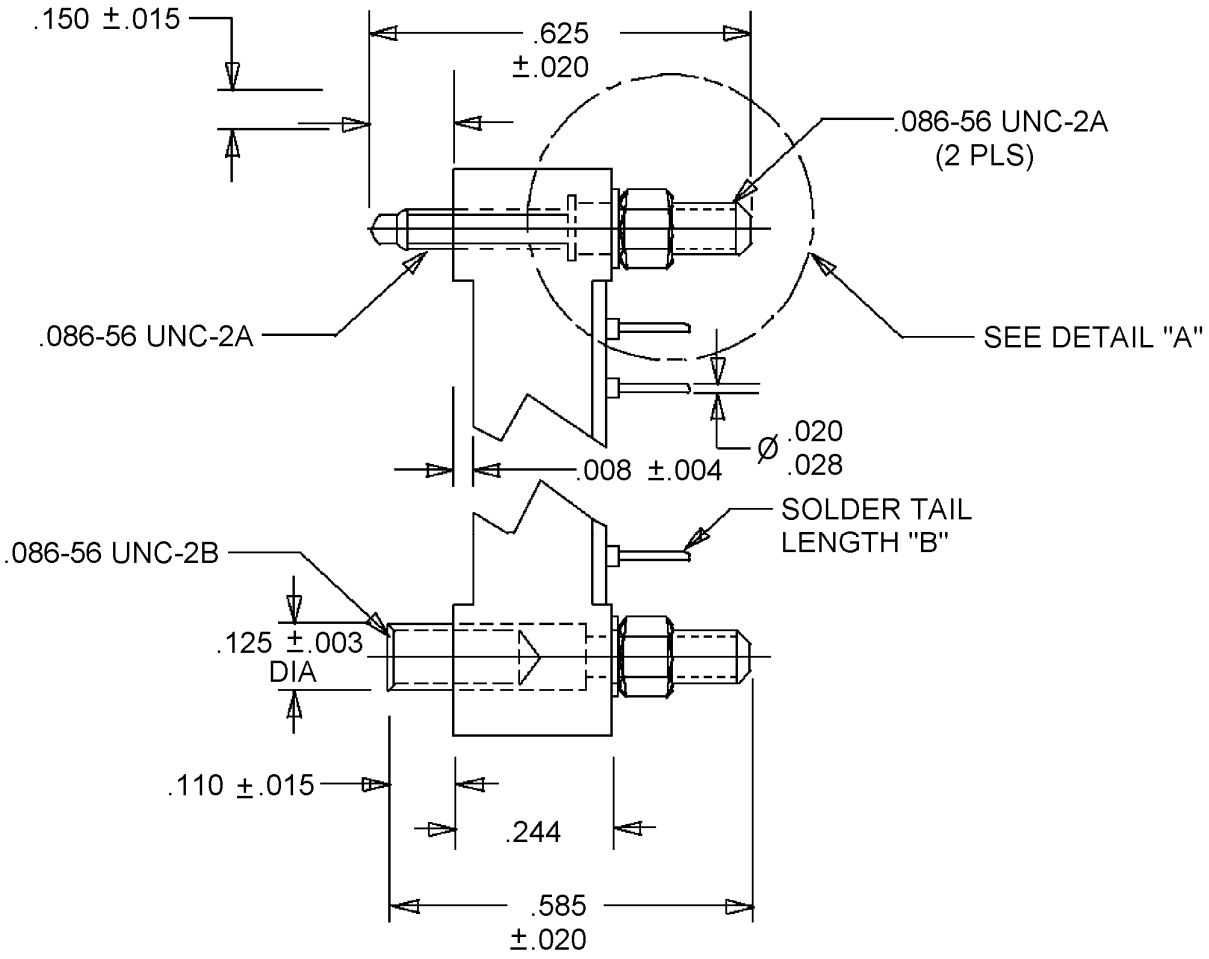

CONNECTOR WITH GUIDEPINS

CONFIGURATION A

FIGURE 1. Connector with jackscrews and guidepins - Continued.



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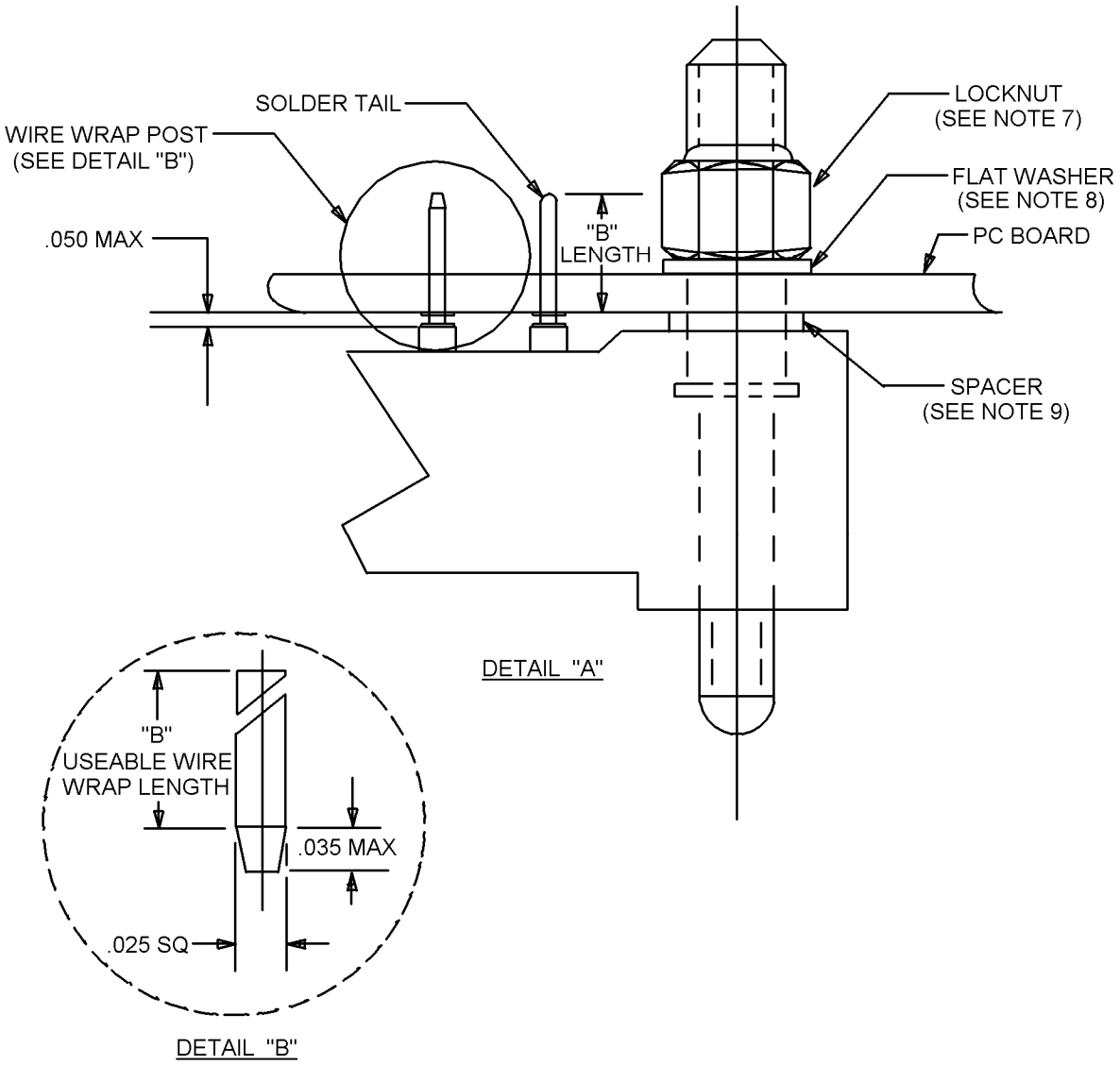

CONNECTOR WITH JACKSCREWS

CONFIGURATION B

FIGURE 1. Connector with jackscrews and guidepins - Continued.



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| Part or identifying Number (PIN) designator | Termination tail style | "B" Length (mm) |
|---|---|---|
| 1A | Straight solder | .093 (2.76) |
| 1B | Straight solder | .125 (3.18) |
| 1C | Straight solder | .156 (3.96) |
| 1D | Straight solder | .187 (4.75) |
| 1E | Wrap post | .260 (6.60) |
| 1F | Wrap post | .390 (9.91) |
| 1G | Wrap post | .550 (13.97) |

FIGURE 1. Connector with jackscrews and guidepins - Continued



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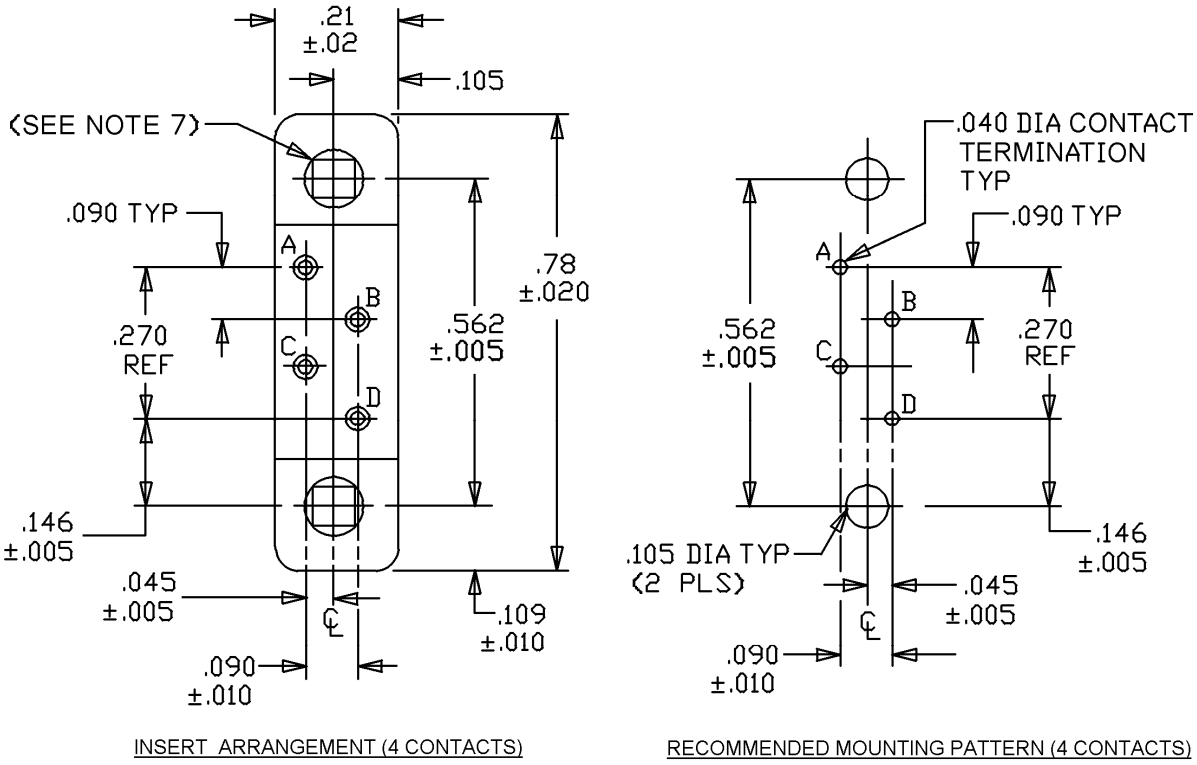
| Inches | mm | Inches | mm |
|---|---|---|---|
| .003 | 0.08 | .050 | 1.27 |
| .004 | 0.10 | .078 | 1.98 |
| .005 | 0.13 | .086 | 2.18 |
| .008 | 0.20 | .090 | 2.29 |
| .01 | 0.25 | .110 | 2.79 |
| .015 | 0.38 | .125 | 3.18 |
| .020 | 0.51 | .150 | 3.81 |
| .025 | 0.63 | .244 | 6.20 |
| .028 | 0.71 | .585 | 14.86 |
| .035 | 0.89 | .625 | 15.88 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances is ± .020 (0.50 mm).
4. Materials for guidepins and jackscrews shall be corrosion resistant steel in accordance with ASTM A276 class 304, 316, or ASTM A582, class 302, 303, 303Se.
5. Material for washer .020 inch (0.51 mm) may be either aluminum or corrosion resistant steel.
6. Finish for washers shall be passivated or anodized.
7. Locknut: Mounting hardware shall include stainless steel locknuts. The nuts may be all metal or designed with nylon inserts.
8. Washer .020 inch (0.51 mm) thick supplied with all sizes.
9. The mounting pad spacer shall be polyamide or vulcanized fiber .040 - 062 inch (1.02 - 1.58 mm) thick.

FIGURE 1. Connector with jackscrews and guidepins - Continued.



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INSERT ARRANGEMENT (4 CONTACTS)

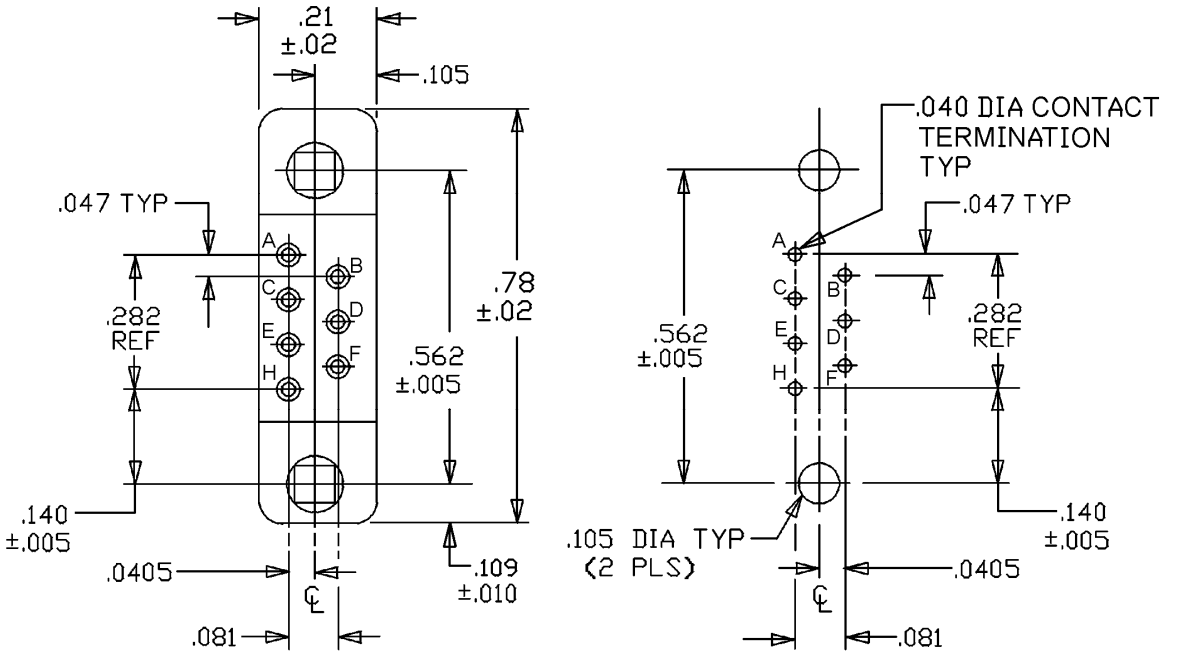
RECOMMENDED MOUNTING PATTERN (4 CONTACTS)

CONFIGURATION A

FIGURE 2. Insert arrangements.



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INSERT ARRANGEMENT (7 CONTACTS)

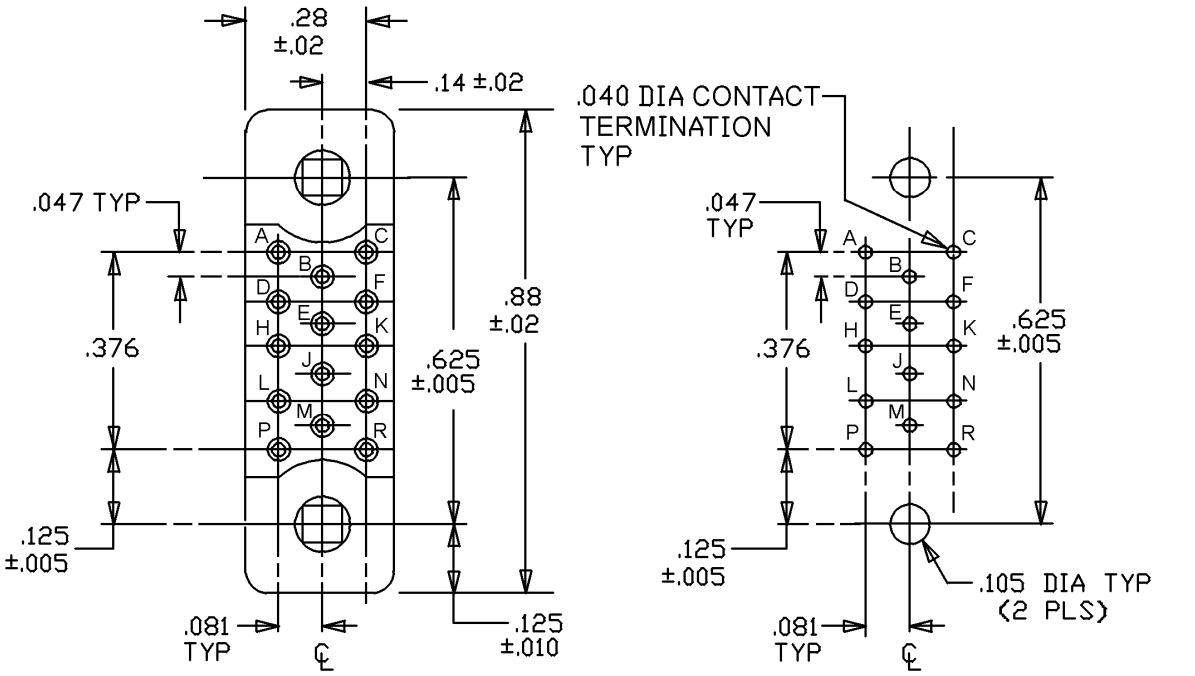
RECOMMENDED MOUNTING PATTERN (7 CONTACTS)

CONFIGURATION B

FIGURE 2. Insert arrangements - Continued.



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INSERT ARRANGEMENT (14 CONTACTS)

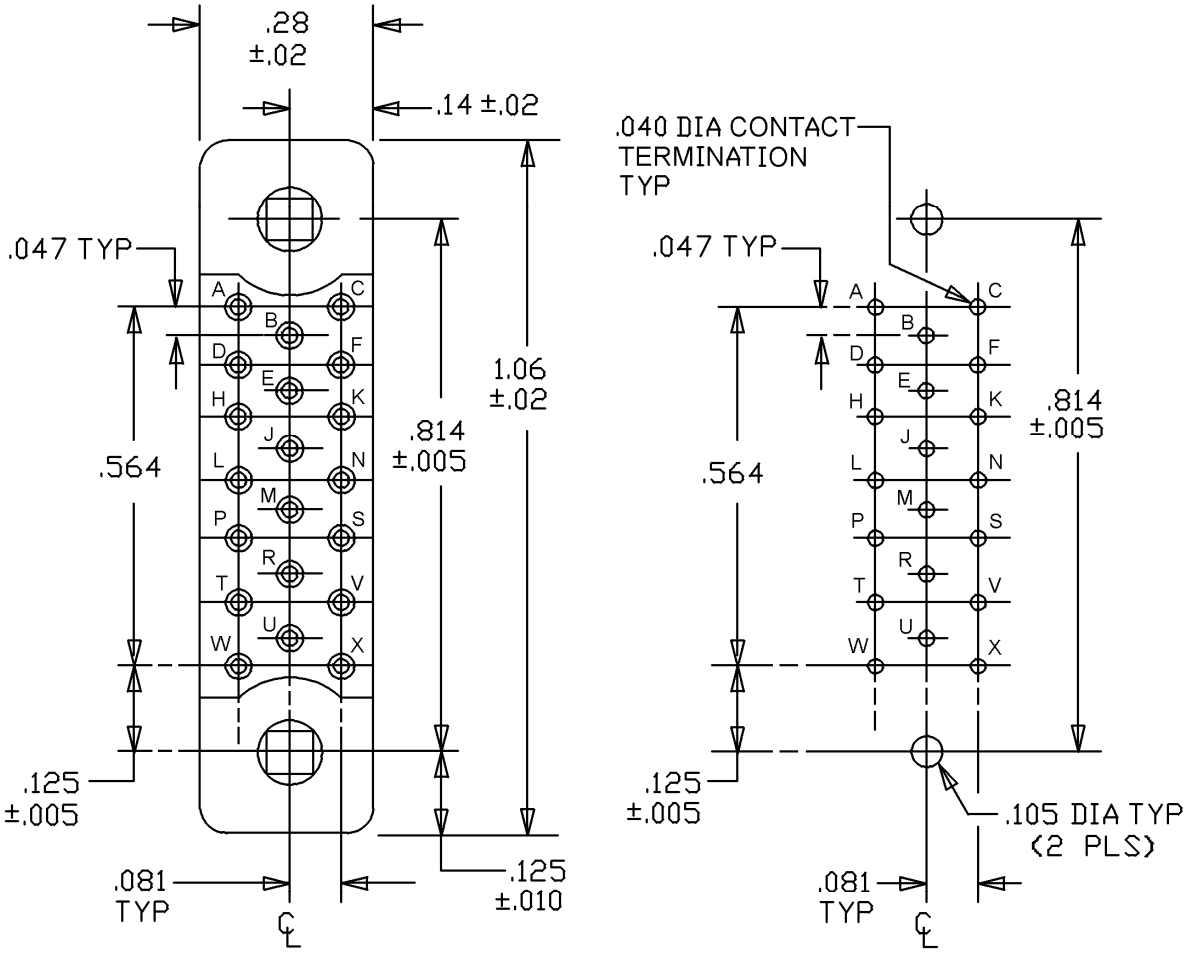
RECOMMENDED MOUNTING PATTERN (14 CONTACTS)

CONFIGURATION C

FIGURE 2. Insert arrangements - Continued.



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INSERT ARRANGEMENT (20 CONTACTS)

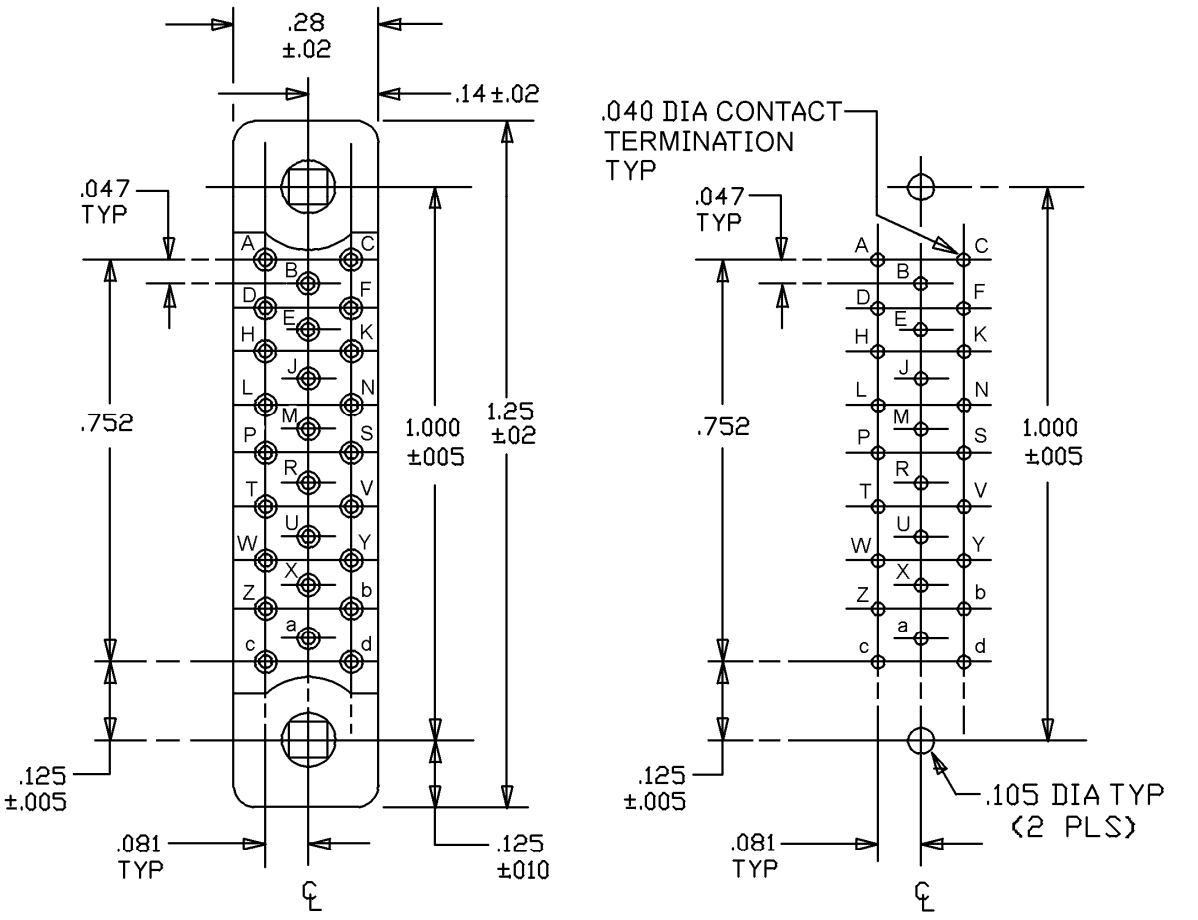
RECOMMENDED MOUNTING PATTERN (20 CONTACTS)

CONFIGURATION D

FIGURE 2. Insert arrangements - Continued.



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INSERT ARRANGEMENT (26 CONTACTS)

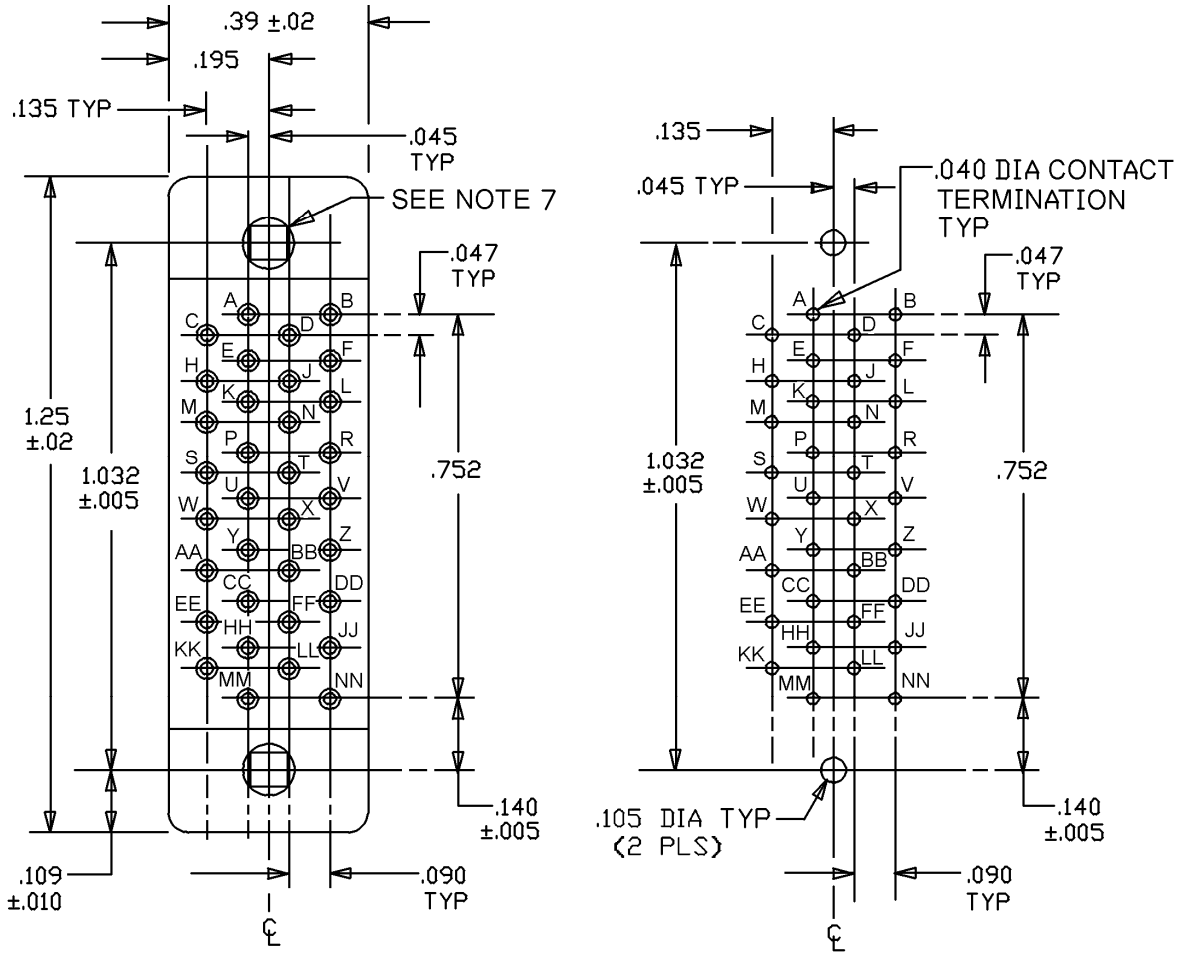
RECOMMENDED MOUNTING PATTERN (26 CONTACTS)

CONFIGURATION E

FIGURE 2. Insert arrangements - Continued.



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INSERT ARRANGEMENT (34 CONTACTS)

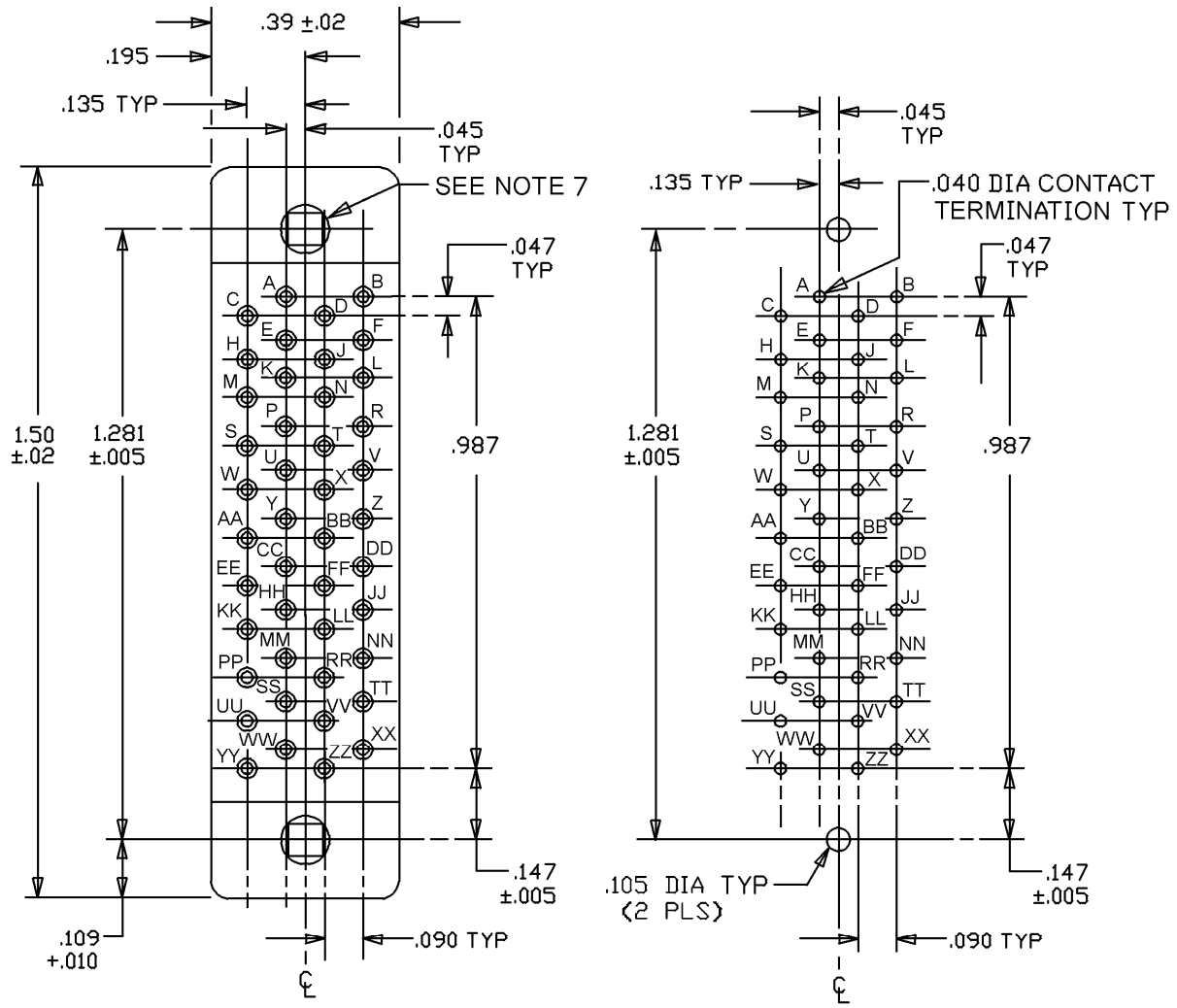
RECOMMENDED MOUNTING PATTERN (34 CONTACTS)

CONFIGURATION F

FIGURE 2. Insert arrangements - Continued.



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INSERT ARRANGEMENT (44 CONTACTS)

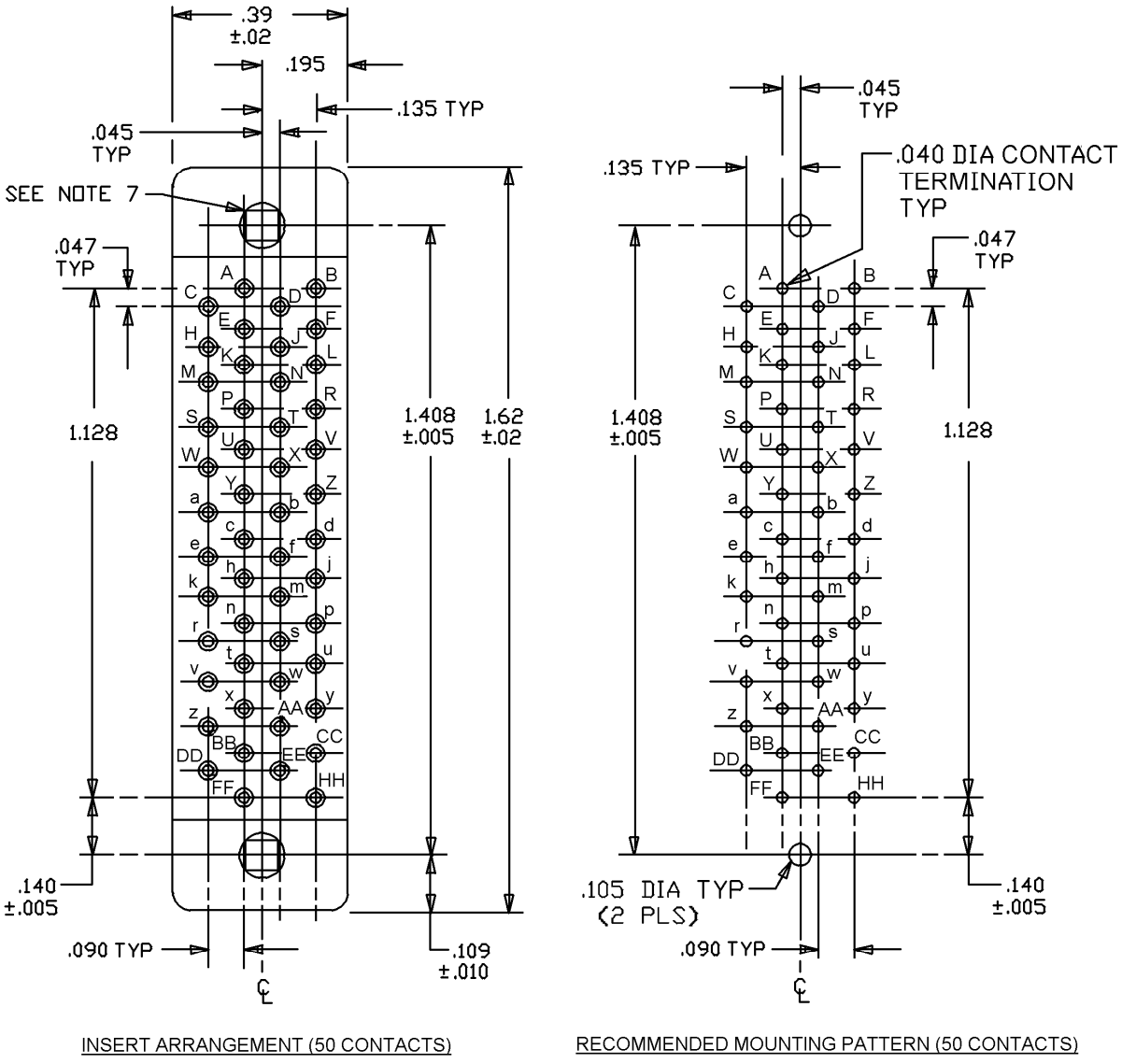
RECOMMENDED MOUNTING PATTERN (44 CONTACTS)

CONFIGURATION G

FIGURE 2. Insert arrangements - Continued.



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INSERT ARRANGEMENT (50 CONTACTS)

RECOMMENDED MOUNTING PATTERN (50 CONTACTS)

CONFIGURATION H

FIGURE 2. Insert arrangements - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:21Z
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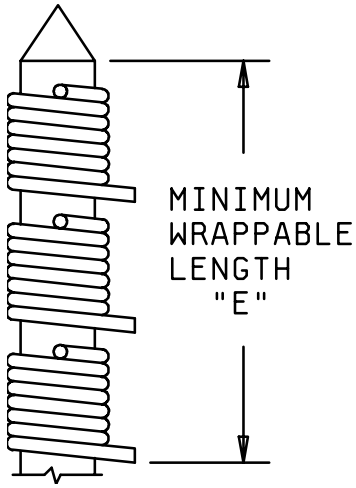
| Inches | mm | Inches | mm |
|---|---|---|---|
| .005 | 0.13 | .28 | 7.1 |
| .010 | 0.25 | .282 | 7.16 |
| .02 | 0.5 | .376 | 9.55 |
| .020 | 0.51 | .39 | 9.91 |
| .040 | 1.02 | .562 | 14.27 |
| .0405 | 1.029 | .564 | 14.33 |
| .045 | 1.14 | .625 | 15.87 |
| .047 | 1.19 | .752 | 19.10 |
| .081 | 2.06 | .78 | 19.8 |
| .090 | 2.29 | .814 | 20.68 |
| .105 | 2.67 | .88 | 22.4 |
| .109 | 2.76 | .987 | 25.06 |
| .125 | 3.18 | 1.000 | 25.40 |
| .135 | 3.43 | 1.032 | 26.21 |
| .14 | 3.6 | 1.06 | 26.9 |
| .140 | 3.56 | 1.128 | 28.65 |
| .146 | 3.71 | 1.25 | 31.8 |
| .147 | 3.73 | 1.281 | 32.54 |
| .195 | 4.95 | 1.408 | 35.76 |
| .21 | 5.3 | 1.50 | 38.1 |
| .270 | 6.86 | 1.62 | 41.1 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerance is ±.015 (0.38 mm).
4. Tolerance between any two adjacent contact centers shall be ±.004 (0.10 mm). Tolerance between any two contact centers, other than adjacent centers, shall be ±.006 (0.15 mm).
5. Insert material in accordance with MIL-DTL-28748.
6. Contact identification may be located on either side of the contact hole but shall appear on the front and rear of each insert.
7. Hole configuration is optional. Design shall incorporate an anti-rotation feature on either front or rear face of insulator.

FIGURE 2. Insert arrangements - Continued.



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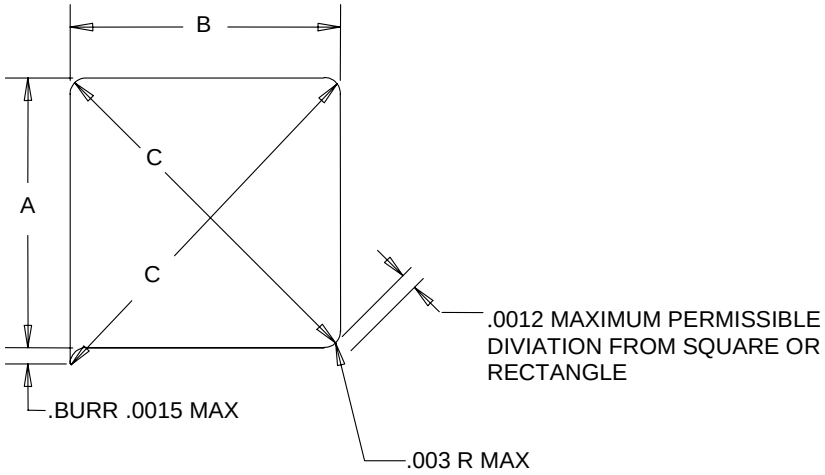

| Number of wrapped corrections | E length | |
|---|---|---|
| | Wire gage 28 (mm) | Wire gage 26 (mm) |
| 1 | .219 (5.56) | .226 (5.74) |
| 2 | .388 (9.85) | .402 (10.21) |
| 3 | .557 (14.15) | .578 (14.68) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Wire wrap posts shall have beveled tips to facilitate wire wrap tools and shall have sharp edges.

FIGURE 3. Wrappable length (typical connection).



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| Inches | mm |
|---|---|
| .0012 | 0.030 |
| .0015 | 0.038 |
| .003 | 0.08 |

| A (mm) | B (mm) | C (mm) | Parallelism (mm) | Straightness Inch/inch (mm) |
|---|---|---|---|---|
| .025 (0.63) nominal<br>.022 (0.56) min | .025 (0.63) nominal<br>.022 (0.56) min | .0355 (0.902) max<br>.0325 (0.825) min | .002<br>(0.05) | .005<br>(0.13) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.

FIGURE 4. Wrappost geometry.



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REQUIREMENTS:

Dimensions and configuration: See figures 1, 2, 3, and 4.

Mating connector: See MIL-DTL-28748/7.

Contacts: Contacts shall be in accordance with MIL-DTL-28748.

Solder tail finish:

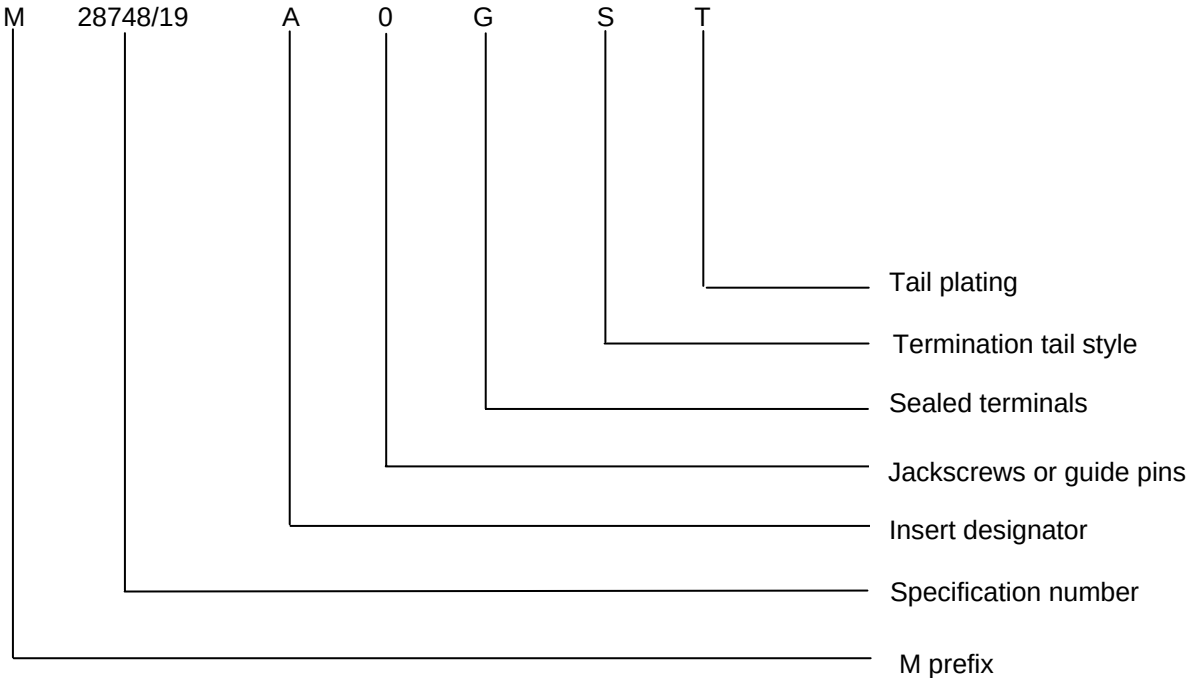
a. Tin/lead in accordance with SAE-AMS-P-81728, 50 to 97 percent tin, 100 microinches (2.54µm) minimum or hot solder dip in accordance with J-STD-006 63 percent tin, 100 micro inches (2.54 µm) minimum over 50 to 100 microinches (1.27 to 2.54µm) of nickel in accordance with SAE-AMS-QQ-N-290, class 1.
b. Gold in accordance with ASTM B488 type II, code C, class 0.76 (30 microinches) min of gold over 50 to 100 microinches (1.27 to 2.54µm) of nickel in accordance with SAE-AMS-QQ-N-290, class 1.

Contact sealants: Sealants used to back seal contact cavities to prevent entry of cleaning solutions, conformal coatings, etc.

Solderless wrappost contacts: Geometry parallelism and tip configuration shall be in accordance with figures 3 and 4.

Solderless wrappost contact wire accommodation: 26 or 28 AWG solid.

Part or Identifying Number (PIN): As shown in the following example: The number zero (0) is used to indicate which parts are not included (see table I permissible PINS).






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| Insert designator | | Number of contact positions |
|---|---|---|
| A | - | 4 |
| B | - | 7 |
| C | - | 14 |
| D | - | 20 |
| E | - | 26 |
| F | - | 34 |
| G | - | 44 |
| H | - | 50 |

Jackscrews or guidepins:

F - fixed jackscrews, configuration B, figure 1.
G - guidepins, configuration A figure 1.
0 - none included.

Jack and guide location: The jackscrew female (turning or fixed) or guide socket shall be located by the A pin contact.

Seal: Manufacturer may supply sealed terminals when no seal is specified.

S - sealed
0 - No seal

Termination tail style:

1A - Straight solder tails .093 inch (2.76 mm).
1B - Straight solder tails .125 inch (3.18 mm).
1C - Straight solder tails .156 inch (3.96 mm).
1D - Straight solder tails .187 inch (4.75 mm).
1E - Wrap post tails .260 inch (6.60 mm).
1F - Wrap post tails .390 inch (9.91 mm).
1G - Wrap post tails .550 inch (13.97 mm).

Tail plating:

T - Tin/lead
G - Gold



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Permissible PINs are specified in table I.

TABLE I. Permissible PINs. 1/

| M prefix and specification number | Insert designator | Jackscrews or guide pins | Sealed terminals | Solder tail plating | Solder tail length |
|---|---|---|---|---|---|
| M28748/19 | A | F - fixed<br>G – Guide pin<br>0 - none | S - sealed<br>0 - no seal | T - tin/lead<br>G - gold | 1A<br>1B<br>1C<br>1D<br>1E<br>1F<br>1G |
| | B | | | | |
| | C | | | | |
| | D | | | | |
| | E | | | | |
| | F | | | | |
| | G | | | | |
| | H | | | | |

1/ The number zero (0) is used to indicate which parts are not included.

Referenced documents: In addition to MIL-DTL-28748, this document references the following:

MIL-DTL-28748/7
ASTM A582
ASTM A276
ASTM B488
J-STD-006
SAE-AMS-P-81728
SAE-AMS -QQ-N-290

CONCLUDING MATERIAL

Custodians:
Army -CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-4672-000)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.



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